

Temperature Control with Arduino

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1 Introduction

Temperature Control is an applicable idea to the regulation of heating and cooling in indoor spaces. “Lab 8C” from Walter F. Smith’s *Experimental Physics: Principles and Practice for the Laboratory* outlines several important concepts that apply temperature control for use with electrical and computerized systems. In this experiment, we explore using a thermistor, a temperature-sensing device that undergoes variation in its resistance with a change in surrounding temperature. The experiment also calls for the use of a chip. Chips combine several switches into one circuit element; each switch can bear a specified amount of current. By providing multiple channels of amplification in a single package, our chip simplifies circuit design and offers protection against back voltage generated by inductive loads. This component is necessary when working with temperature control systems because currents and voltages may fluctuate at potentially high values to accomplish desired temperature shifts.

Two main types of temperature control are attempted in our experiment: “Bang-Bang Control” and “Proportional Control.” Bang-Bang operates under a simple “if-else” statement. If the current temperature sensed by the thermistor is lower than the desired value, the heating element is turned on. The heating element turns off once the desired value is exceeded. This is a binary output (on or off) and results in the heater completely engaging and disengaging numerous times to maintain the set temperature over time. Proportional Control attempts to eliminate the binary nature of Bang-Bang. Instead of the system switching completely on or off, it increases or decreases the voltage supplied to the heating element depending on how far the current temperature is from the desired. The response of the system is proportional to how much error, or how far the temperature is from what is set. This method utilizes a “gain” which dictates how much error will be tolerated. The higher the gain, the more strict the Proportional Control is about error. Because the heater is no longer switched totally on and off, power is saved and we can sustain a set temperature in a more calculated manner.

Our objective with this experiment is to understand Bang-Bang and Proportional Control methods by comparing them in practice and considering their real-world applications.

2 Procedure

The lab introduces the many different working parts of temperature control systems in steps. Below is a list of components used for the experiment:

- Thermistor
- Arduino Uno circuit board and accompanying software
- Heating Element
- DC power supply
- ULN2803 chip
- Breadboard to connect components

To start, we test the thermistor coupled with the Arduino Uno Board and software. We adapt the code provided by Sunfounder, the company that runs the Arduino Uno board, for this experiment. The code contains the steps for setting up the LCD display, which we did not use because it was an extra moving part that we did not want to risk causing issues for little to no benefit to our project. To ensure an accurate reading from the thermistor, we used a clip to keep it in solid contact with the heating element. In the software, Arduino IDE, we apply the code and add the designated libraries. In order to see the output, we press the serial output button in the top right corner of the window to show us the output values. Once we have a functioning fundamental circuit, we then make the full circuit, shown in **Figure 1**, which we will use to test the different temperature control methods: Bang-Bang and Proportional Control.

The defining factor between these two methods is the code the software runs. The Bang-Bang utilizes an “if-else” statement, which turns pin 9 on the Arduino board to high voltage, allowing the heater to receive current. This statement is shown in **Figure 2**.

The serial output in Arduino IDE displays the temperature sensed by the thermistor once each second. We take data for about 5 minutes, allowing the temperature to fluctuate above and below our desired value of 85°F, and create the plot in Microsoft Excel. We repeat the same process for Proportional Control, adapting the Arduino IDE code. Instead of a binary output, we now have an analog output. This code is shown in **Figure 3**.

3 Results

We use Excel’s Data Streamer to create a chart for each method, showcased in **Figure 4** and **Figure 5**.

The success of each method is assessed by determining the average fluctuation in temperature and the χ^2 value of each. The average fluctuation is the mean value of the amplitude of each node in the plot. This presents us a general idea of how effective the heating technique was at maintaining a consistent

temperature. The χ^2 value tests how agreeable the measured temperatures are with our "expected" value of 85°F. The χ^2 formula is given by

$$\chi^2 = \sum_{i=1}^n \frac{(Observed - Expected)^2}{Expected}$$

Our resulting numbers are shown below.

Control Method	Average Amplitude	Chi-Squared
Bang-Bang Control	.68125	.165972
Proportional Control	.62222	.142108

We conclude that Proportional Control, due to its lower variability, is a more efficient temperature control method. More advanced Proportional Control methods are the modern heating/cooling industry standard. If more work is to be done on this experiment, it would be worth looking into the power consumption of each method and perhaps having multiple temperature sensors: placed at key locations within the room being heated or cooled.

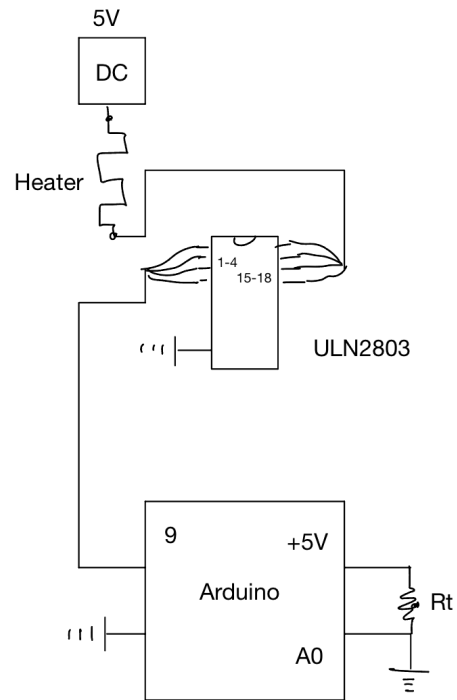


Figure 1: Full circuit design

```

49
50  if (TempF < 85){
51    digitalWrite(9,HIGH)
52  }
53  else {
54    digitalWrite(9,LOW)
55  }
56

```

Figure 2: Bang-Bang Control code

```

56
57  float error = 85 - tempF;
58  int response = int (500 * error);
59  if (response>255) response = 255;
60  if (response<1) response = 0;
61  analogWrite (9,response);
62  }

```

Figure 3: Proportional Control code

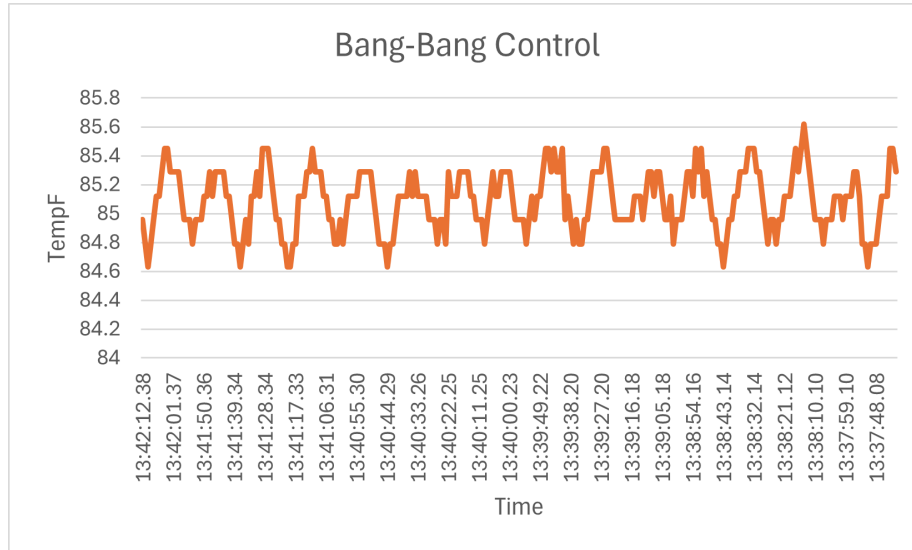


Figure 4: Temperature in Fahrenheit versus Time graph of Bang-Bang Control

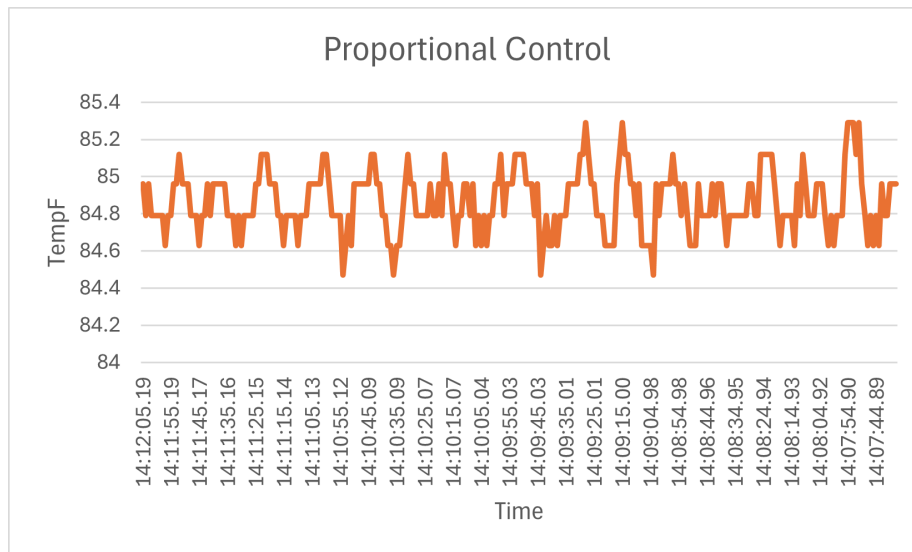


Figure 5: Temperature in Fahrenheit versus Time graph of Proportional Control